Part III: Coxeter Groups

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1 Definition and Examples

1.1 Abstract Definition

Definition 1.1. Let X be a set, the free group on X is defined to be the set $\text{Free}(X)$ of words in $x \in X$ and x^{-1} . Group operation is the concatenation of words.

Let G be a group, a presentation of G is a set X of generators, and a set $R \subset \text{Free}(X)$ of relations, together with an isomorphism $\text{Free}(X)/N(R) \simeq G$, where $N(R)$ is the smallest normal subgroup of $\text{Free}(X)$ containing R. We write:

$$G = \langle x \in X \mid r = 1, \forall r \in R \rangle = \langle X \mid R \rangle$$

Example. *Let $G = \langle X \mid R \rangle, H = \langle Y \mid S \rangle$, then*

$$G \times H = \langle X \coprod Y \mid R, S, xy = yx, \forall x \in X, y \in Y \rangle$$

Definition 1.2. Let I be any set, $X = \{x_i : i \in I\}$. A **Coxeter matrix** is a $|I| \times |I|$ symmetric matrix $M = (m_{ij})_{i,j \in I}$ such that $m_{ij} \in \mathbb{N} \cup \{\infty\}$, and $m_{ii} = 1, m_{ij} \geq 2, \forall i \neq j$.

A **Coxeter system** is a triple (W, I, M) for M Coxeter matrix over I, and

$$W = \langle x_i, i \in I \mid (x_i x_j)^{m_{ij}} = 1, \forall i, j \in I \rangle$$

We call $|I|$ the rank of the system, and W a Coxeter group.

Remark. *1. For all $i \in I$, $x_i^2 = 1$*

2. $m_{ij} = 2 \iff x_i x_j = x_j x_i$

Example. *Let I be any set, $M = (m_{ij})$ for $m_{ij} = \begin{cases} 1, & \text{if } i = j; \\ 2, & \text{if } i \neq j \end{cases}$. Then $W = \bigoplus_I \mathbb{Z}/2\mathbb{Z}$.*

These are the only abelian Coxeter groups.

1.2 Coxeter Graph

Fix (W, I, M) Coxeter system.

Definition 1.3. The Coxeter graph of (W, I, M), denoted by $\Gamma(W)$, is the following labeled graph:

1. vertices: elements $i \in I$
2. edges: $i, j \in I$ are joined by edge if $m_{ij} > 2$
3. labelling: if $m_{ij} > 3$, then the edge between i, j is labeled by m_{ij}

1.3 First Examples

1. $I = \{1\}, W = \mathbb{Z}/2\mathbb{Z}$. So $\Gamma(W) = \bullet$. More generally, if I is any set, $W = \bigoplus_I \mathbb{Z}/2\mathbb{Z}$, $\Gamma(W)$ is $|I|$ disjoint vertices.

2. Dihedral Groups. Let $I = \{1, 2\}$. For al $3 \leq n \leq \infty$, let $M_n = \begin{pmatrix} 1 & n \\ n & 1 \end{pmatrix}$

   (a) If $n < \infty$, $W = \langle x_1, x_2 \mid x_1^2 = x_2^2 = (x_1x_2)^n = 1\rangle = \langle x_1, x_1x_2\rangle \simeq D_{2n}$

   (b) If $n = \infty$, $W = \langle x_1, x_2 \mid x_1^2 = x_2^2\rangle = D_\infty$

$$\Gamma(W) = \begin{cases} \bullet\!\!-\!\!\bullet, & \text{if } n = 3; \\ \bullet\overset{n}{-}\bullet, & \text{otherwise.} \end{cases}$$

3. The symmetric group. Let $X = \{1, 2, \cdots, n\}$, $S_n = \mathrm{Sym}(X)$ generated by transpositions $s_i = (i, i+1), 1 \leq i \leq n-1$. Note: $(i, j) = (j, j-1)(i, j-1)(j, j-1)$, and transpositions generate S_n. Let

$$m_{ij} = \begin{cases} 3, & \text{if } |i-j| = 1; \\ 2, & \text{if } |i-j| > 1; \\ 1, & \text{otherwise.} \end{cases}$$

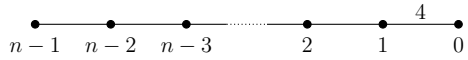
   and $W = \langle x_1, \cdots, x_{n-1} \mid (x_ix_j)^{m_{ij}} = 1\rangle$, $\Gamma(W) =$ is of type A_{n-1}.

   Note there's a surjection $W \twoheadrightarrow S_n$. We then compute cardinality of left hand side to see this is an isomorphism.

4. The signed symmetric group.

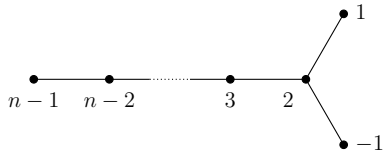
   Let $X = \{\pm 1, \pm 2, \cdots, \pm n\}$, $SS_n = \{p \in \mathrm{Sym}(X) \mid p(-i) = -p(i)\}$. Note that

$$SS_n = \langle s_0 = (1, -1), s_i = (i, i+1)(-i, -i-1) : i < n\rangle$$

   is a Coxeter group. $\Gamma(SS_n) =$  is of type B_n

5. The even signed symmetric group.

$$ESS_n = \{p \in SS_n \mid \#\{i > 0 : p(i) < 0\} \text{ is even}\} = \langle s_{-1} = (1, -2)(-1, 2), s_1, \cdots, s_{n-1}\rangle$$

$\Gamma(ESS_n) =$ 

2 Reflection Groups

2.1 Isometries and Reflections

Let V be a finite dimensional inner product space over $\mathbb{R}$. Note V is a complete metric space under the classical Euclidean metric. For all $T \in \mathrm{End}(V)$, it has a unique adjoint map T^*, corresponding to taking transpose on the level of matrices. We can thus define $\mathrm{O}(V)$ the group of orthogonal transformations.

Definition 2.1. A map $\alpha : V \to V$ is isometry (not necessarily linear) if for any $u, v \in V$, $d(\alpha u, \alpha v) = d(u, v)$. Note if T is liner isometry $\iff T \in \mathrm{O}(V)$.

Proposition 2.2. *An isometry $\alpha : V \to V$ is linear $\iff \alpha(0) = 0$*

Proof. One direction is trivial. We assume $\alpha(0) = 0$. Since α preserves distance and angles, we can easily see $\alpha(u+v) = \alpha(u)+\alpha(v)$. Moreover, for the same reason we see $\alpha(\lambda u) = \mu\alpha(u)$ for $|\lambda| = |\mu|$. Now if $\lambda = -\mu$, then

$$\|\lambda u - u\| = \|= \alpha(\lambda u) - \alpha(u)\| = \|-\lambda u - u\|$$

So $|\lambda - 1| = |\lambda + 1| \implies \lambda = 0$. Contradiction. So $\lambda = \mu$. □

Let $\mathcal{I}(V)$ be the group of isometries, $\mathrm{O}(V) \subset \mathcal{I}(V)$. For each $v \in V$, let T_v be the translation by v. $T(V) = \{T_v \mid v \in V\} \leq \mathcal{I}(V)$. Clearly, $T(V) \simeq (V, +)$.

Corollary 2.3. *$\mathcal{I}(V)$ is generated by $\mathrm{O}(V)$ and $T(V)$, $\mathrm{O}(V) \cap T(V) = \{\mathrm{id}\}$.*

Remark. *An isometry α has a fixed point $\iff \alpha = T\beta T^{-1}$ for $T \in T(V), \beta = \mathrm{O}(V)$.*

Definition 2.4. A hyperplane in V is a linear subspace of V of codimension 1. An affine hyperplane is a subset of form $T(H)$ for $T \in T(V)$, H hyperplane. An isometry $\alpha \in \mathcal{I}(V)$ is a reflection if $H_\alpha \coloneqq \{v \in V \mid \alpha(v) = v\}$ is an affine hyperplane.

2.2 Standard Reflections

Let V be any real vector space, B any symmetric bilinear form on V.

Definition 2.5.

1. B nondegenerate $\iff V^\perp = 0$
2. B positive semidefinite $\iff B(u, u) \geq 0, \forall u \in V$
3. B positive definite $\iff B(u, u) \geq 0$ and it equals 0 if and only if $u = 0$.

Definition 2.6. Given $v \in V, B(v, v) \neq 0$. The standard reflection at v is

$$s_v : V \to V, u \mapsto u - \frac{2B(u, v)}{B(v, v)}v$$

Lemma 2.7. *Given $v \in V, B(v, v) \neq 0$.*

1. *$s_v^2 = \mathrm{id}$, $s_v(v) = -v$, $s_{\lambda v} = s_v$ for any $\lambda \neq 0$.*

2. $H_v = \langle v \rangle^\perp$ *is a hyperplane.*

3. $s_v \in \mathrm{O}(V; B)$

4. *If* $T \in \mathrm{O}(V; B)$, *then* $T s_v T^{-1} = s_{Tv}$

Now we assume $\dim V < \infty$ and B inner product.

Proposition 2.8. *$\alpha \in \mathcal{I}(V)$ is a reflection iff $\alpha = T s_v T^{-1}$ for some $T \in T(V), v \in V$.*

Proof. Clearly if α is of the given form, then α is a reflection. Now we prove the converse. Set $H_\alpha = T(H)$, then $\beta = T^{-1}\alpha T$ is a linear reflection. Hence without loss of generality, we may assume $\alpha = s$ is a linear reflection.

In this case $H_s = \{v \in V \mid s(v) = v\}$ is a hyperplane. Choose an orthogonal basis $\{e_1, \cdots, e_n\}$ such that $\{e_1, \cdots, e_{n-1}\}$ is a basis for H_s. Let $v = e_n$. We claim $s(e_i) = s_v(e_i)$. Indeed, this trivially holds $i < n$. For e_n, note if $s(e_n) = \lambda_1 e_1 + \cdots \lambda_n e_n \implies 0 = (e_i, e_n) = (s(e_i), s(e_n)) = \lambda_i$ for each $i < n$. So $s(e_n) = \lambda_n e_n$. So $\|s(e_n)\| = \|e_n\| \implies \lambda_n = \pm 1$. But if $\lambda_n = 1 \implies s = \mathrm{id}$. Contradiction. So $s(e_n) = -e_n = s_v(e_n)$. Hence $s = s_v$. □

2.3 Reflection Groups and Root Systems

Let $(V, (\cdot, \cdot))$ be finite dimensional inner product space.

Definition 2.9. A reflection group in V is a subgroup of $\mathcal{I}(V)$ generated by reflections. A linear reflection group (LRG) is a subgroup of $\mathrm{O}(V)$ generated by linear reflections. A finite reflection group is a LRG which is finite.

Example. 1. $\mathrm{O}(V)$ *is a LRG and* $\mathcal{I}(V)$ *is a reflection group, but* $T(V)$ *is not.*

2. *Let* $V = \mathbb{R}$, $\mathrm{O}(V) = \{\pm \mathrm{id}\}$ *is a FRG. Now if* $s = -\mathrm{id}$ *reflection about* 0, $t = T_{\frac{1}{2}} \circ s \circ T_{-\frac{1}{2}}$ *reflection around* $\frac{1}{2}$. *Then* $D_\infty = \langle s, t \rangle \leq \mathcal{I}(\mathbb{R})$ *is a reflection group.*

3. *Let* $V = \mathbb{R}^2$, *and* Γ *be a regular* n*-gon. Then* $\mathrm{Sym}\Gamma = \{\alpha \in \mathcal{I}(V) \mid \alpha(\Gamma) = \Gamma\} \simeq D_{2n}$ *is a FRG.*

4. *Let* $V = \mathbb{R}^n$, *a convex* n*-polyhedron* Γ *in* V *is the boundary of a convex bounded intersection of the set of half spaces. Then* Γ *is union of faces, and* $\mathrm{Sym}(\Gamma)$ *acts on the set of faces, and each face is a* $(n-1)$*-convex polyhedron.*

   We say Γ *is regular, if* $\mathrm{Sym}(\Gamma)$ *acts transitively on the faces. In fact if* $V = \mathbb{R}^3$, Γ *is regular polyhedron iff:*

   (a) *All faces are similar*

   (b) *Each face is a regular polygon*

   (c) *Each vertex meets the same number of faces*

   In fact, they are classified as ***Platonic solids****:*

   (a) Γ_4*:* 4 *faces (triangle),* 6 *edges,* 4 *vertices (tetrahedron)*

(b) Γ_6*: 6 faces (square), 12 edges, 8 vertices (cube)*

(c) Γ_8*: 8 faces (triangle), 12 edges, 6 vertices (octahedron)*

(d) Γ_{12}*: 12 faces (pantagon), 30 edges, 20 vertices (dodecahedron)*

(e) Γ_{20}*: 20 faces (triangle), 30 edges, 12 vertices (icosahedron)*

We see $\mathrm{Sym}(\Gamma_4) \simeq S_4$*;* $\mathrm{Sym}(\Delta_n) = S_n$*;* $\mathrm{Sym}(\Gamma_6) = \mathrm{Sym}(\Gamma_8) \simeq SS_3$*. In fact, the symmetry of n-hypercube is* SS_n*. Moreover,* $\mathrm{Sym}\Gamma_{12} = \mathrm{Sym}\Gamma_{20}$*. They are all FRG.*

Definition 2.10. A nonempty subset $\Phi \subset V$ with $|\Phi| < \infty, 0 \notin \Phi$ is a **root system** if

1. If $\alpha \in \Phi$, then $\Phi \cup R\alpha = \{\pm\alpha\}$
2. For all $\alpha \in \Phi$, $s_\alpha(\Phi) = \Phi$.

   We call the elements of Φ roots. We say Φ is crystallographic if in addition:
3. for all $\alpha, \beta \in \Phi$, $\langle \alpha, \beta \rangle = \frac{2(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Z}$.

Definition 2.11. If Φ is a root system, then the reflection group of Φ is defined to be

$$W(\Phi) = \langle s_\alpha \mid \alpha \in \Phi \rangle \subset \mathrm{O}(V)$$

This is a LRG. If Φ is crystallographic, then call it a Weyl group.

Theorem 2.12. *1. For any root system* Φ*,* $W(\Phi)$ *is a FRG.*

2. If W *is any FRG, ther exists a root system* Φ *such that* $W = W(\Phi)$*.*

Proof. 1. Let $\theta : W(\Phi) \to \mathrm{Sym}(\Phi)$ be the action map. We will see θ is injective. Now if $w_0 \in W(\Phi)$ such that $\theta(w_0) = 1$, then w_0 fixes each root. Let $U = \mathrm{Span}\Phi \leq V$, then w_0 fixes U pointwise. Now $V = U \oplus U^\perp$. If $v \in U^\perp$, then for all $\alpha \in \Phi$, $s_\alpha(v) = v, \forall \alpha \in \Phi \implies w_0(v) = v$. So w_0 also fixes $U^\perp$ pointwise. So $w_0 = \mathrm{id}$.

2. Let $\Phi = \{v \in V \mid \|v\| = 1, s_v \in W\}$. One easily sees Φ is finite, nonempty, and $0 \notin \Phi$. Moreover $W = W(\Phi)$ since its a finite reflection group. We need to check the root system axioms. But both are straightforward.

□

Example. *1. Let* $V = \mathbb{R}$*,* $\Phi = \{\pm\alpha\}$*,* $W(\Phi) = \mathbb{Z}/2\mathbb{Z}$*.*

2. Let $V = \mathbb{R}^2$*,* $\alpha \in V$*,* $\theta = \frac{2\pi}{n}$*. Let* r *be rotation by* θ *and* $\Phi = \{r^m(\alpha) \mid m \in \mathbb{Z}\}$*. This is a root system, with reflection group* D_{2n}*.*

2.4 Fundamental Sysmtem

Definition 2.13. A subset $\Delta \subset \Phi$ is a fundamental system if

1. Δ is basis for $U = \text{Span}\Phi$
2. $\Phi = \mathbb{Z}_{\geq 0} \cdot \Delta \coprod \mathbb{Z}_{\leq 0} \cdot \Delta$

Elements of Δ are call simple roots.

Now fix a basis $B = \{\alpha_1, \cdots, \alpha_n\} \subset \Phi$ for U. Let

$$\Pi = \{\alpha \in \Phi \mid \alpha = \sum_{i=N}^{n} b_i \alpha_i, b_i > 0\}$$

Note this set is closed under addition, and $\Phi = \Pi \coprod -\Pi$. We call Π a positive system. Let $\Delta \subset \Pi$ be the indecomposable roots.

Lemma 2.14. *For all $\alpha, \beta \in \Delta$ with $\alpha \neq \beta$, we have $(\alpha, \beta) \leq 0$.*

Proof. Suppose $(\alpha, \beta) > 0$, then $s_\alpha(\beta) = \beta - \frac{2(\alpha,\beta)}{(\alpha,\alpha)}\alpha \in \Phi$. Let $c = \frac{2(\alpha,\beta)}{(\alpha,\alpha)} > 0$. So $\beta - c\alpha \in \Pi$ or $c\alpha - \beta \in \Pi$. In either case, there exists $\alpha, \beta \in \Delta$ such that $a\alpha - b\beta \in \Pi$ for $a, b > 0$. Namely

$$a\alpha - b\beta = \sum_{\gamma \in \Delta} c_\gamma \cdot \gamma, c_\gamma \geq 0$$

1. If $c_\alpha < a$, then
$$(a - c_\alpha)\alpha = b\beta + \sum_{\gamma \in \Delta \setminus \{\alpha\}} c_\gamma \cdot \gamma$$
Hence cwe can discard α from Δ, contradicting the minimality of Δ.

2. If $c_\alpha \geq a$, then
$$0 = (c_\alpha - a)\alpha + b\beta + \sum_{\gamma \neq \alpha} c_\gamma \cdot \gamma = \sum_{\gamma \in \Delta} d_\gamma \cdot \gamma$$
for $d_\gamma \geq 0$ for all γ. But this is clearly a contradiction if we write the above expression in coordinates B.

Therefore, we must have $(\alpha, \beta) \leq 0$. □

Theorem 2.15. *Δ is a fundamental system.*

Proof. Since $\Phi = \Pi \coprod -\Pi$, so any $\alpha \in \Phi$ can be written as $\sum_{\gamma \in \Delta} c_\gamma \cdot \gamma$ for c_γ of the same signs. Clearly, Δ spans U. It remains to show Δ is a linearly independent set.

Assume $0 = \sum c_\gamma \cdot \gamma$ for $c_\gamma \neq 0$ for some γ. Let $\Delta_1 = \{\gamma \in \Delta \mid c_\gamma > 0\}, \Delta_2 = \{\gamma \in \Delta \mid c_\gamma < 0\}$. For each $\beta \in \Delta_2$, set $d_\beta = -c_\beta > 0$. So

$$\sum_{\alpha \in \Delta_1} c_\alpha \cdot \alpha = \sum_{\beta \in \Delta_2} d_\beta \cdot \beta = \sigma$$

By looking at (σ, σ) we see $\sigma = 0$. So $\sum_{\alpha \in \Delta_1} c_\alpha \cdot \alpha = 0$. But this is again a contradiction if we write it out using B coordinates. □

In summary, we proved:

1. Fundemental system exists for each Φ
2. For any two elements in the fundamental system, their inner product is ≤ 0.

Definition 2.16. For a fundamental system $\Delta \subset \Phi$, the Dynkin diagram of Δ is the graph $\Gamma(\Delta)$ with multiple edges defined by:

1. Vertices: elements $\alpha \in \Delta$
2. Edges: If $\alpha, \beta \in \Delta, \alpha \neq \beta$. If the angle between α, β is of form $\pi - \frac{1}{n}\pi$, then they are connected by $(n-2)$ edges.

2.5 The Reflection Group

Let $\Phi \subset V$ be a root system, $\Delta \subset \Phi$ be a fundamental system, Π a positive system.

Lemma 2.17. *For all $\alpha \in \Delta$, $s_\alpha(\Pi \backslash \{\alpha\}) = \Pi \backslash \{\alpha\}$*

Proof. Let $\beta \in \Pi \backslash \{\alpha\}$ and write $\beta = \sum_{\gamma \in \Delta} c_\gamma \cdot \gamma$, $c_\gamma \geq 0$ and there exists $\gamma \neq \alpha$ such that $c_\gamma > 0$. We compute

$$s_\alpha(\beta) = \sum_{\gamma \neq \alpha} c_\gamma s_\alpha(\gamma) - c_\alpha \cdot \alpha = \sum_{\gamma \neq \alpha} c_\gamma \cdot \gamma + k\alpha$$

for some $k \in \mathbb{Z}$. This forces $s_\alpha(\beta) \in \Pi$. It is also evident that $s_\alpha(\beta) \neq \alpha$. □

Theorem 2.18. $W(\Phi) = \langle s_\delta : \delta \in \Delta \rangle$

Proof. Let W' be the subgroup generated by simple reflections. We will prove for all $\beta \in \Pi$, there exists $\alpha \in \Delta$ and $w \in W'$ such that $w(\alpha) = \beta$. Note this implies $s_\beta \in W'$ and we are done. Hence we need to prove $W' \cdot \beta \cap \Delta \neq \emptyset$.

Clearly, $\beta \in W'\beta \cap \Pi \neq \emptyset$. Let $m := \min\{\sum_{\gamma \in \Delta} c_\gamma \mid \sum_{\gamma \in \Delta} c_\gamma \cdot \gamma \in W'\beta \cap \Pi\} > 0$. Choose $\alpha \in W' \cdot \beta \cap \Pi$ such that $ht(\alpha) = m$. Note $0 < (\alpha, \alpha) = \sum_{\gamma \in \Delta} c_\gamma(\gamma, \alpha)$. So $(\delta, \alpha) > 0$ for some $\delta \in \Delta$.

Now if $\delta = \alpha$, then $\alpha \in \Delta$ we're done. So we assume $\delta \neq \alpha \implies \alpha \in \Pi \backslash \{\delta\}$. So $s_\delta(\alpha) \in \Pi \backslash \{\delta\}$. But $s_\delta \in W' \implies s_\delta(\alpha) \in W' \cap \Pi$. But $ht(s_\delta(\alpha)) < ht(\alpha)$, contradiction. So $W'\beta \cap \Delta \neq \emptyset$. □

Let Φ be a root system, let $m_{\alpha,\beta}$ be the order of $s_\alpha s_\beta \in W(\Phi)$ for $\alpha, \beta \in \Delta$. Let $M_\Delta = (m_{\alpha,\beta})$ be the corresponding Coxeter matrix, and $(\widetilde{W(\Delta)}, \Delta, M_\Delta)$ be the associated Coxeter system. There exists a a surjection

$$\widetilde{W(\Delta)} \to W(\Phi)$$

Proposition 2.19. *The Coxeter graph of $(\widetilde{W(\Delta)}, \Delta, M_\Delta)$ is equivalent to the Dynkin diagram of Δ.*

3 Coxeter Groups and Representations

Fix a Coxeter system (W, I, M).

3.1 Geometric Representations

Definition 3.1. The associated space of M is the $\mathbb{R}$-vector space $V_M = \text{Span}\{e_i \mid i \in I\}$. Define the associated bilinear form to be

$$(\cdot,\cdot)_M : V_M \times V_M \to \mathbb{R}, (e_i, e_j)_M = \begin{cases} \cos(\pi - \frac{\pi}{m_{ij}}), & \text{if } m_{ij} < \infty; \\ -1, & \text{otherwise.} \end{cases}$$

This is a symmetric bilinear form on V_M.

Remark. *Even if I is finite, $(\cdot,\cdot)_M$ may not be positive semidefinite. But if $(\alpha,\alpha)_M \neq 0$, we can still s_α.*

Definition 3.2. The geometric representation of Coxeter group W is the group homomorphism

$$\sigma_W : W \to \text{GL}(V_M), x_i \mapsto s_{e_i}$$

Define $\Phi_M = \{\sigma_W(w)(e_i) \mid i \in I, w \in W\}$.

Proposition 3.3. *If I is finite, W is finite, and $(\cdot,\cdot)_M$ is an inner product. Then Φ_M is a root system in V_M.*

Proof. One easily checks Φ_M is nonempty and finite. Moreover, $0 \neq \Phi_M$. Given $\alpha, \beta \in \Phi_M$, with $\alpha = \sigma(w)(e_i), \beta = \sigma(w')(e_j)$.

1. $\Phi_M \cap \mathbb{R}\alpha = \{\pm\alpha\}$. If $\beta \in \Phi_M \cap \mathbb{R}\alpha$, $\beta = \lambda\alpha$. Then $\lambda\sigma(w')(e_j) = \sigma(w)e_i$. So $\lambda e_j = \sigma(w'^{-1}w)(e_i)$. So we know $(\lambda e_j, \lambda e_j) = (e_i, e_i) \implies \lambda = \pm 1$. Moreover, $-\alpha = \sigma_W(s_{e_i}(e_i)) \implies -\alpha \in \Phi_M$.

2. $s_\alpha(\beta) \in \Phi_M$. Indeed, we compute

   $$s_\alpha(\beta) = s_{\sigma(w)(e_i)}(\beta) = \sigma(w)s_{e_i}\sigma(w)^{-1}(\beta) = \sigma(wx_iw^{-1})\sigma(w')(e_j) = \sigma(wx_iw^{-1}w')(e_j)$$

   This lies in Φ_M.

□

3.2 the Length Formula

Let (W, I, M) be a Coxeter system, with generators $x_i, i \in I$.

Definition 3.4. Let $w \in W$. The lenght of w, denoted as $l(w)$, is the minimal length of expressions of w using words $\{x_i, i \in I\}$.

Lemma 3.5. *Let $w, w' \in W, i \in I$.*

1. $l(w) = 0 \iff w = 1$
2. $l(w) = l(w^{-1})$
3. $l(ww') \leq l(w) + l(w')$
4. $l(w) - 1 \leq l(wx_i) \leq l(w) + 1$

Hence we can define $d(w, w') = l(ww')$ to be a metric on W. This gives W the discrete topology.

Proposition 3.6. *There exists a unique homomorphism*

$$\varepsilon : W \to \mathbb{Z}/2\mathbb{Z}, w \mapsto l(w) (\text{mod } 2)$$

Example. 1. *$W = D_{2n}$, $\ker \varepsilon$ are the rotatoins.*
2. *$W = D_\infty$, $\ker \varepsilon$ are the translations.*
3. *$W = S_n$, $\ker \varepsilon = A_n$.*

Corollary 3.7. *If $w \in W, i \in I$, then $l(wx_i) = l(w) \pm 1, l(x_i w) = l(w) \pm 1$.*

3.3 Faithful Representations

Let $(V_M, (\cdot, \cdot))$ be the associated space, and $\sigma = \sigma_W : W \to \mathrm{GL}(V_M)$ be the geometric representation.

Example. *Let $I = \{1, 2\}$, $V_M = \mathbb{R}^2$, $M = \begin{pmatrix} 1 & n \\ n & 1 \end{pmatrix}$. If $n < \infty$, then $\sigma(x_1 x_2) = s_{e_1} s_{e_2}$ is rotation by $\frac{2\pi}{n}$.*

If $v \in V_M$, we say $v > 0$ if $v = \sum_{i \in I} \alpha_i e_i$ with each $\alpha_i > 0$. For $w \in W, v \in V_M$, we denote $w \cdot v = \sigma(w)(v)$.

Theorem 3.8. 1. *For all $w \in W, i \in I$, if $l(wx_i) > l(w)$, then $w \cdot e_i > 0$*
2. *For all $w \in W, i \in I$, if $l(wx_i) < l(w)$, then $w \cdot e_i < 0$*
3. *$\sigma : W \to \mathrm{GL}(V_M)$ is faithful.*

Proof. If $|I| \leq 2$, this is an exercise. Now assume $|I| \geq 3$. Given $i, j \in I$, define $W_{ij} = \langle x_i, x_j \rangle \leq W$. Let $\widetilde{W_{ij}} = \langle y_i, y_j \mid y_i^2 = y_j^2 = (y_i y_j)^{m_{ij}} = 1 \rangle$ be a Coxeter group. We clearly have a surjection $\pi : \widetilde{W_{i,j}} \to W_{i,j}$. Moreover, we know $\sigma_{\widetilde{W_{ij}}} = \sigma_W |_{W_{ij}} \circ \pi$. But by the case for $|I| = 2$, we know $\sigma_{\widetilde{W_{ij}}}$ is faithful. So π is injective. Hence $\pi : \widetilde{W_{ij}} \simeq W_{i,j}$. So W_{ij} is a Coxeter group with length function l_{ij} such that $l_{ij}(w) \geq l(w)$ for all $w \in W_{ij}$.

We now see that that parts 2 and 3 follows from the first part. For the third part, suppose $w \neq 1 \in W$ with $\sigma(w) = \mathrm{id}$. There exists x_i such that $l(wx_i) < l(w)$. So we

know $e_i = w \cdot e_i < 0$, contradiction. For the second part, assume $l(wx_i) < l(w)$. Then $l(wx_i \cdot x_i) > l(wx_i) \implies wx_i \cdot e_i > 0 \implies we_i < 0$.

Now it remains to prove the first part. We perform induction on $l(w)$. The case for $l(w) = 0$ is trivial.

Now suppose $l(w) = n > 0$ with $l(wx_i) > l(w)$. Choose $j \in I$ such that $l(wx_j) < l(w)$. Let $W_{ij} = \langle x_i, x_j \rangle$ be the Coxeter group of length function l_{ij}. Define

$$A = \{w' \in W \mid w'^{-1}w \in W_{ij}, l(w) = l(w') + l_{ij}(w'^{-1}w)\}$$

Note that $wx_j \in A$, so A is nonempty. Choose $w_0 \in A$ such that $l(w_0)$ is minimal. So $l(w_0) \leq l(wx_j) = l(w) - 1$. Set $v = w_0^{-1}w \in W_{ij}$. So $w = w_0 v$, $l(w) = l(w_0) + l_{ij}(v)$.
Claim: $l_{ij}(vx_i) > l_{ij}(v), l(w_0x_i) > l(w_0), l(w_0x_j) > l(w_0)$

It is clear that the result follows from this claim.
Proof of Claim:

1. Suppose $l_{ij}(vx_i) < l_{ij}(v) \implies l(wx_i) = l(w_0vx_i) \leq l(w_0)+l_{ij}(vx_i) = l(w_0)+l_{ij}(v)-1 < l(w)$. This is a contradiction. So $l_{ij}(vx_i) > l_{ij}(v)$

2. For $k \in \{i, j\}$, suppose $l(w_0x_k) = l(w_0) - 1$. So $l(w) = l(w_0v) = l(w_0x_k \cdot x_kv) \leq l(w_0x_k) + l_{ij}(x_kv) = l(w_0) - 1 + l_{ij}(x_kv) \leq l(w_0) + l_{ij}(v) = l(w)$. So in fact the equality holds through. So indeed $w_0x_k \in A$, contradicting the minimality. So $l(w_0x_k) = l(w_0) + 1$.

□

Let V be a finite dimensional inner product space. $\Phi \subset V$ root system, $\Delta \subset \Phi$ fundamental system, $W(\Phi) = \langle s_\alpha, \alpha \in \Phi \rangle$. Let $m_{\alpha,\beta}$ be the order of $s_\alpha s_\beta, \alpha, \beta \in \Delta$, $M_\Delta = (m_{\alpha,\beta})_{\alpha,\beta\in\Delta}$.

Corollary 3.9. *$(W(\Phi), \Delta, M_\Delta)$ is a Coxeter system. In particular, all FRGs are Coxeter groups.*

Proof. We know $W(\Phi) = \langle s_\delta \mid \delta \in \Delta \rangle$. For each $\delta \in \Delta$, let $e_\delta = \frac{\delta}{\|\delta\|}$ and they form a basis of V. We now have the geometric representation $\widetilde{W(\Delta)}$:

$$\widetilde{\sigma} : \widetilde{W(\Delta)} \to \mathrm{GL}(V)$$

with image $W(\Phi)$. Moreover, $\widetilde{\sigma}$ is faithful so in fact $\widetilde{W(\Delta)} \simeq W(\Phi)$. □

Corollary 3.10. *If $|W| < \infty$, $(\cdot, \cdot)_M$ is positive definite, then W is a FRG.*

Proof. Consider geometric representation

$$\sigma : W \to \mathrm{GL}(V)$$

with image $W(\Phi_M)$. But by faithfulness $W \simeq W(\Phi_M)$ is a FRG. □

3.4 Exchange Condition and Deletion Condition

Theorem 3.11 (exchange condition)**.** *Let $w = x_{i_1} \cdots x_{i_r}$, and suppose $k \in I$ such that $l(wx_k) < l(w)$. Then there exists **unique** $j \in \{1, 2, \cdots, r\}$ such that $wx_k = x_{i_1} \cdots \widehat{x_{i_j}} \cdots x_{i_r}$*

Proof. For convenience, set x_j to be x_{i_j}. Since $l(wx_k) < l(w) \implies w \cdot e_k < 0$. But $e_k > 0$, so there exists j such that $x_{j+1} \cdots x_r \cdot e_k > 0$ but $x_j \cdots x_r \cdot e_k < 0$.

Let $w' = x_{j+1} \cdots x_r$. So $\beta = w'e_k > 0$, $x_j\beta < 0$. We write out the expression for β:

$$\beta = \sum_{i \in I} \alpha_i e_i, \alpha_i \geq 0$$

So $x_j\beta = s_{e_j}(\beta) = \beta - 2(\beta, e_j)e_j = \sum_{i \neq j} \alpha_i e_i + (\alpha_j - 2(e_j, \beta))e_j < 0$. So we must have $\alpha_i = 0$ for $i \neq j$ and $\beta = \alpha_j e_j$. But note $(\beta, \beta)_M = (e_k, e_k)_M = 1$. So we must have $\beta = e_j$. So $w'e_k = e_j$. We can now compute:

$$\sigma(w'x_k) = \sigma(w')\sigma(x_k)\sigma(w'^{-1})\sigma(w') = s_{\sigma(w')e_k}\sigma(w') = s_{e_j}\sigma(w') = \sigma(x_j w')$$

By faithfulness of σ, $w'x_k = x_j w'$. So we know $wx_k = x_1 \cdots x_j w' x_k = x_1 \cdots x_{j-1} w' = x_1 \cdots \widehat{x_j} \cdots x_r$.

To see uniqueness, suppose $wx_k = x_1 \cdots \widehat{x_j} \cdots x_r = x_1 \cdots \widehat{x_m} \cdots x_r$ for $j < m$. We note

$$x_{j+1} \cdots x_m = x_j \cdots x_{m-1}$$

Then $w = x_1 \cdots x_j \cdot x_j \cdots x_{m-1} \cdots x_r = x_1 \cdots \widehat{x_j} \cdots \widehat{x_m} \cdots x_r$ has length $\leq r - 2$. Contradiction. □

Corollary 3.12 (deletion condition)**.** *The following results hold.*

1. *If $w = x_{i_1} \cdots x_{i_r}$ for $l(w) < r$, then there exists $k < j$ such that*

$$w = x_{i_1} \cdots \widehat{x_{i_j}} \cdots \widehat{x_{i_k}} \cdots x_{i_r}$$

2. *A reduced expression of w can be obtained by removing successive pairs of generators.*

Proof. Let $j \geq 0$ be maximal such that $l(w_j) = j$, where $w_j = x_{i_1} \cdots x_{i_j}$. We know $l(w_{j+1}) = j - 1$. So by exchange condition, we know

$$x_{i_1} \cdots x_{i_{j+1}} = x_{i_1} \cdots \widehat{x_{i_k}} \cdots x_{i_j}$$

for some k. □

3.5 Braid Relations

Note given $i, j \in I$ and assume $x_i x_j$ has order m_{ij}, then

$$\underbrace{x_i x_j \cdots}_{m_{ij} \text{ times}} = \underbrace{x_j x_i \cdots}_{m_{ij} \text{ times}}$$

We say $(i_1, \cdots, i_r) \sim_{\text{b.e.}} (j_1, \cdots, j_r)$ are related by **braid relations** if: they give the same element in W and there exists a sequence of braid relations connecting these two.

Example. *Let $m_{ij} = 3, m_{jk} = 2$, then $(i, j, i, k) \sim (j, i, k, j)$.*

Theorem 3.13 (Matsumoto's Theorem)**.** *If $l(w) = r$, then any two reduced expressions of w are related by braid relations.*

Proof. Let $w = x_{i_1} \cdots x_{i_r} = x_{j_1} \cdots x_{j_r}$. We prove they're related by braid relations by induction on r. The case for $r = 0, 1$ is trivial. Now if $i_r = j_r$, then we are done by induction hypothesis. So we may assume $i_r \neq j_r$. So

$$l(wx_{i_r}) = l(w_{j_r}) = l(w) - 1$$

Now choose s maximal such that

$$w = x_{m_1} \cdots x_{m_{r-s}} \underbrace{\cdots x_i x_j x_i}_{s}$$

where $i, j \in \{i_r, j_r\}, i \neq j$. Note that $l(wx_j) < l(w)$, so there are three possible cases:

1. $wx_j = x_{m_1} \cdots \widehat{x_{m_k}} \cdots x_{m_{r-s}} \cdots x_i x_j x_i$, so $w = x_{m_1} \cdots \widehat{x_{m_k}} \cdots x_{m_{r-s}} \underbrace{\cdots x_i x_j}_{s+1}$, contradicting the maximality of s.
2. $wx_j = x_{m_1} \cdots x_{m_{r-s}} \underbrace{\cdots x_j x_i x_j}_{s-1}$. Then $w = x_{m_1} \cdots x_{m_{r-s}} \underbrace{\cdots x_j x_i}_{s-2} \implies l(w) \leq r - 2$, contradiction.
3. $wx_j = x_{m_1} \cdots x_{m_{r-s}} \underbrace{\cdots x_i x_j x_i}_{s-1} \implies w = x_{m_1} \cdots x_{m_{r-s}} \underbrace{\cdots x_j x_i x_j}_{s}$. Compare with the initial expression we see $(x_i x_j)^s = 1 \implies m_{ij} \mid s$. In fact $s = m_{ij}$ since we started with a reduced word.

Now we may assume $i = i_r, j = j_r$. Then we have:

$$(i_1, \cdots, i_r) \sim (m_1 \cdots, m_{r-s}, \cdots, i, j, i) \sim (m_1 \cdots, m_{r-s}, \cdots, j, i, j) \sim (j_1, \cdots, j_r)$$

by induction. We're done. □

Corollary 3.14. *If $\Gamma(W, I, M)$ contains a cycle, then W is infinite.*

Proof. Suppose $(i_1, \cdots, i_n)$ is a cycle of minimal length. Let $x_j = x_{i_j}$. For each $k \in \{1, \cdots, n\}$, we set

$$W_i^{(k)} = x_k \cdot x_{k+1} \cdots x_{k+i-1}$$

where we take the index to be modulo n. Now if W is finite, we can find j minimal such that

$$l(W_j^{(k)}) < j$$

for some k. Without loss we assume $k = 1$ and set $w_i = w_i^{(1)}$. So we know $l(w_{j-1}) = j - 1, l(w_j) = j - 2$. By exchange condition, we know

$$w_{j-1}x_j = x_1 \cdots \widehat{x_k} \cdots x_{j-1} \iff x_{k+1} \cdots x_{j-1} = x_{k-1} \cdots x_1 w_{j-1} x_j$$

If $k > 1$, we compute

$$W_{j-k}^{(k+1)} x_j = x_{k+1} \cdots x_{j-1} = x_{k-1} \cdots x_1 w_{j-1} x_j$$

So $l(W_{j-k}^{(k+1)}) \leq j-k-1$, contradicting the minimality of k. So $k = 1$, and $w_{j-1}x_j = x_2 \cdots x_{j-1}$ is the reduced word. Hence $x_{1 \cdots j-1} = x_2 \cdots x_j$ as reduced words. Apply Matsumoto, we see

$$(1, 2, \cdots, j-1) \sim (2, 3, \cdots, j)$$

But these two expressions cannot be related by braid relations. Contradiction. □

4 Inner Product and Finite Coxeter Groups

Let (W, I, M) be a Coxeter system. We say it is irreducible if its Coxeter graph is connected. From now on we assume I is finite and $I = I_1 \coprod \cdots \coprod I_r$ for each I_j a connected component of $\Gamma(W)$. We set $W_j = \langle x_i \mid i \in I_j \rangle$. Then W_j is an irreducible Coxeter group, and $W = W_1 \times \cdots \times W_r$. Let $\sigma : W \to \mathrm{GL}(V_M)$ be the geometric representation.

4.1 Subrepresentations of V_M

If $I = I_1 \coprod \cdots \coprod I_r$, then set $V_j = \mathrm{Span}\{e_j \mid i \in I_j\}$, $V = V_1 \oplus \cdots \oplus V_r$.

Lemma 4.1. *Each V_j is a subsrepresentation of V. W_k acts trivially on V_j if $k \neq j$, and V_j is the geometric representation of W_j.*

Lemma 4.2. *(V, σ) is completely reducible $\iff$ each V_j is completely reducible.*

Proposition 4.3. *1. $V^\perp$ is a W-subrepresentation of V, and $V = V_1^\perp \oplus \cdots V_r^\perp$*

2. If (W, I, M) is irreducible, then any proper subrepresentation of V is contained $V^\perp$

3. V is completely reducible $\iff$ $(\cdot, \cdot)_M$ is nondegenerate.

4. W finite $\implies$ $(\cdot, \cdot)_M$ nondegenerate.

Proof. 1. This is clear.

2. Suppose $U \leq V$ proper subrepresentation. Given $i \in I$, note $x_i u \in U \implies u - 2(e_i, u)e_i \in U \implies$ either $e_i \in U$ or $(e_i, u) = 0$ for all $u \in U$.

   Now if $e_i \notin U$ for each $i \in I$, then $U \leq V^\perp$, we're done. So we assume $e_i \in U$ for some i. But by the connectedness of $\Gamma(W)$ we say $e_i \in U$ for each $i \implies U = V$. Contradiction.

3. V completely reducible $\iff$ V_j completely reducible for each $j \iff V_j^\perp = 0$ for each $j \iff (\cdot, \cdot)_M$ is nondegenerate.

4. Maschke's theorem.

□

Lemma 4.4. *If $\phi \in \mathrm{End}_{\mathbb{R}}(V)$ and ϕ commutes with W-actions, then ϕ is a scalar map.*

Proof. Note $-\phi(e_i) = \phi(x_i(e_i)) = x_i\phi(e_i)$. But x_i is standard reflection with -1-eigenspace spanned by e_i. So we know $\phi(e_i) = \lambda_i e_i$ for some $\lambda_i \in \mathbb{R}$. Fix i and choose $\lambda = \lambda_i$, then $\phi - \lambda \cdot \mathrm{id}$ has a nontrivial kernel. Suppose this is not everything, then $\ker(\phi - \lambda \cdot \mathrm{id}) \subset V^{\perp}$, contradicting the fact that $e_i \in \ker(\phi - \lambda \cdot \mathrm{id})$. □

Proposition 4.5. *Suppose W irreducible, $B : V \times V \to \mathbb{R}$ is symmetric nondegenerate W-invariant bilinear form. If in addition $(\cdot,\cdot)_M$ is nondegenerate, then $B = \lambda(\cdot,\cdot)_M$ for some $\lambda \neq 0$.*

Proof. Clear □

Proposition 4.6. *If W is finite, then $(\cdot,\cdot)$ is positive definite.*

Proof. We only need to construct one positive definite W-invariant symmetric bilinear form. Indeed, this can be done using a standard averaging argument. □

Corollary 4.7. *A finite Coxeter group is a FRG.*

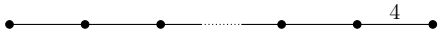
Remark. *In fact,* $\{FRG\} \longleftrightarrow \{$*finite Coxeter group*$\} \longleftrightarrow \{$*positive Coxeter group*$\}$

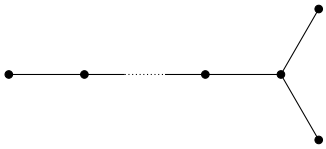
4.2 Classification of Coxeter Groups of Positive Type

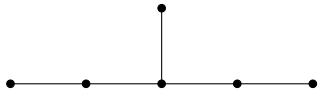
Definition 4.8. Let (W, I, M) be a Coxeter system. We call it has positive type if $(\cdot,\cdot)_M$ is positive semidefinite; it is positive if $(\cdot,\cdot)_M$ is positive definite.

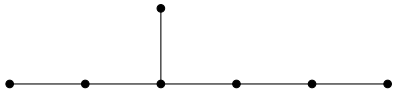
We have the following classification of irreducible positive Coxeter groups:

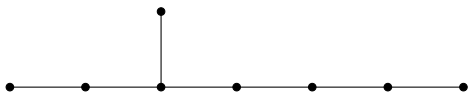
$A_n (n \geq 1)$: $, W = S_{n+1}$

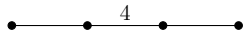
$B_n (n \geq 2)$:  $, W = SS_n$

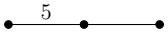
$D_n (n \geq 4)$:  $, W = ESS_n$

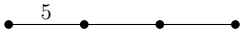
E_6 : 

E_7 : 

E_8 : 

F_4 : 

H_3 : 

H_4 : 

$I_2(m)(m \geq 5)$: 

Note we have

$$I_2(3) = A(2), I_2(4) = B_2, I_2(5) = H_2, I_2(6) = G_2$$

For rank 3 groups, A_3 is the symmetry of tetrahedron, B_3 is the symmetry of cube, H_3 is the symmetry of icosahedron.

Note if $I = \{1, 2, \cdots, n\}$, $V = \mathbb{R}^n$. (W, I, M) is irreducible $\iff$ A is indecompasble (i.e. there doesn't exist $J, K \subset I$ such that $a_{jk} = 0$ for all $j \in J, k \in K$)

Proposition 4.9. *If A symmetric, indecompasble, $a_{ij} \leq 0$ for $i \neq j$. Assume A is positive semidefinite. Then*

$$N = \{v \in V \mid Av = 0\} = \{v \in V \mid v^t Av = 0\}$$

Moreover, either $N = 0$ or $N = \langle z \rangle$ such that $z_i > 0$ for all i.

Proof. Exercise. □

Corollary 4.10. *If W is irreducible of positive type, and for all $j \in I$ set $W_j = \langle x_i \mid i \neq j \rangle$. Then W_j is a positive Coxeter group. (i.e. remove any vertex from $\Gamma(W)$, we get a disjoint union of "positive diagrams").*

Proof. Consider N be defined as in the previous proposition. It has dimension 0 or 1. If $N = 0$ then the result is clear. If $N = \langle \alpha_1 e_1 + \cdots + \alpha_n e_n \rangle$ for all $\alpha_i > 0$. Then removing any e_i will kill N. □

4.3 The Coxeter Complex

Let $(V, (\cdot, \cdot))$ be a finite dimensional inner product space. Let $\mathcal{H}$ be a set of (not necessarily linear) hyperplanes in V. Denote s_H as reflection through $H \in \mathcal{H}$.

More concretely, we can write $H = T(H_0), T = T_{\alpha_H} \in T(V)$, $H_0 = \{v \in V \mid (v, u_H) = 0\}$. So

$$H = \{v \in V \mid (v, u_H) = (\alpha_H, u_H) = \lambda_H\}$$

We can divide V into half spaces:

$$A_{1,H} = \{v \in V \mid (v, u_H) > \lambda_H\}; \quad A_{-1,H} = \{v \in V \mid (v, u_H) < \lambda_H\}$$

So $V = H \coprod A_{1,H} \coprod A_{-1,H}$. Note we may assume $\|u_H\| = 1$, and if we swap the sign of u_H, we swap the two half spaces.

Now let $W(\mathcal{H}) = \langle s_H \mid H \in \mathcal{H} \rangle \subset \mathcal{I}(V)$ be the reflection group of $\mathcal{H}$. Without loss we may assume $\mathcal{H}$ is $W(\mathcal{H})$-invariant.

Definition 4.11. 1. We say $\mathcal{H}$ is **locally finite** if there exists $R > 0$ such that for each $v \in V$, the ball $B(v, R)$ intersects only finitely many $H \in \mathcal{H}$.

2. We say $W \subset \mathcal{I}(V)$ is an **affine reflection group** if $W = W(\mathcal{H})$ for some locally finite, W-invariant set of affine hyperplanes.

Fix $W = W(\mathcal{H})$ affine reflection group. Set $X = V \backslash \bigcup_{H \in \mathcal{H}} H$. There is a natural equivalence relation on X: $x \sim y \iff$ for each H x and y lie in the same half space.

Let $\mathcal{C} = X/\sim$ be the **Coxeter Complex**. We call an element of $\mathcal{C}$ a **chamber**. For each chamber $C \in \mathcal{C}$, $H \in \mathcal{H}$, we say H is a **wall** of C if $\overline{C} \cup H = \overline{U}$ for U open in H. Denote $\mathrm{Wall}(C)$ be the collection of walls of C.

Given $C, D \in \mathcal{C}, C \neq D$, we say C and D are **adjacent** if there exists $H \in \mathrm{Wall}(C) \cap \mathrm{Wall}(D), \overline{C} \cap H = \overline{D} \cap H$. We write $C \sim_H D$.

Example. $V = \mathbb{R}, \mathcal{H} = \frac{1}{2}\mathbb{Z}, W(\mathcal{H}) = D_\infty$.

Proposition 4.12. *1. For each $C \in \mathcal{C}$, $\mathrm{Wall}(C)$ is a finite set, C is a finite intersection of half spaces corresponding to hyperplanes in $\mathrm{Wall}(C)$. In particular, all chambers are open.*

2. For each $C \in \mathcal{C}, H \in \mathrm{Wall}(\mathcal{C})$, there exists a unique chamber D adjacent to it via H.

3. For any $C, D \in \mathcal{C}$, there exists a gallery

$$C = C_0 \sim C_1 \sim C_2 \sim \cdots \sim C_m = D$$

Proof. We only prove the first statement. Fix a chamber $C \in \mathcal{C}$. Let

$$\mathcal{H}_1 = \{H \in \mathcal{H} \mid C \subset A_{1,H}\}, \quad \mathcal{H}_{-1} = \{H \in \mathcal{H} \mid C \subset A_{-1,H}\}$$

So $C = \bigcap_{H \in \mathcal{H}_1} A_{1,H} \cap \bigcap_{H \in \mathcal{H}_{-1}} A_{-1,H}$, and $\mathcal{H}_1 \coprod \mathcal{H}_2 = \mathcal{H}$.

Note if $H, H' \in \mathcal{H}_1$ with $u_H = u_{H'}$. Then if $\lambda_H > \lambda_{H'}$, then $A_{1,H} \subset A_{1,H'}$. So if we fix $H \in \mathcal{H}_1$, we denote $\lambda^{(H)} \coloneqq \sup\{\lambda_{H'} \mid H' \in \mathcal{H}_1, u_{H'} = u_H\} < \infty$, since C is nonempty. Moreover, $\lambda^{(H)} = \lambda_{H'}$ for some $H' \in \mathcal{H}$ with $u_{H'} = u_H$. Since if there exists a sequence $\lambda_{H_i} \to \lambda^{(H)}$ but not eventually constant, then it contradicts locally finiteness.

Now let's denote $\mathcal{H}_1' \coloneqq \{H \in \mathcal{H}_1 \mid \lambda_H = \max\{\lambda_{H'} : u_H = u_{H'}\}\}$,

$$\mathcal{H}_{-1}' \coloneqq \{H \in \mathcal{H}_{-1} \mid \lambda_H = \min\{\lambda_{H'} : u_H = u_{H'}\}\}$$

Then $C = \bigcap_{H \in \mathcal{H}_1'} A_{1,H} \cap \bigcap_{H \in \mathcal{H}_{-1}'} A_{-1,H}$. But $\{u_H \mid H \in \mathcal{H}\}$ is a finite set by locally finiteness. Hence $\mathcal{H}_1', \mathcal{H}_2'$ are finite sets.

Assume $H \in \mathrm{Wall}(C) \cap \mathcal{H}_1$. If $\lambda_H < \lambda_{H'}$ for some $H' \in \mathcal{H}_1$ with $u_H = u_{H'}$. We write $\lambda_{H'} = \lambda_H + \varepsilon$. Hence for every $v \in C$, we have

$$(v, u_H) = (v, u_{H'}) > \lambda_{H'} = \lambda_H + \varepsilon$$

This shows $\overline{C} \cap H = \emptyset$, contradiciton. Hence in fact $H \in \mathcal{H}_1' \implies \mathrm{Wall}(C) \cap \mathcal{H}_1 \subset \mathcal{H}_1'$. Similarly, we have $\mathrm{Wall}(C) \cap \mathcal{H}_{-1} \subset \mathcal{H}_{-1}'$. So indeed $\mathrm{Wall}(C)$ is a finite set. Moreover

$$C \subset \bigcap_{H \in \mathrm{Wall}(C) \cap \mathcal{H}_1} A_{1,H} \cap \bigcap_{H \in \mathrm{Wall}(C) \cap \mathcal{H}_{-1}} A_{-1,H} \subset \bigcap_{H \in \mathcal{H}_1'} A_{1,H} \cap \bigcap_{H \in \mathcal{H}_{-1}'} A_{-1,H} = C$$

So equality goes through. □

Remark. *In fact, such an intersection is the unique "minimal" intersection. Indeed, assume*

$$C = A_{1,H_1} \cap \cdots A_{-1,H_m}$$

$H_i \in \mathcal{H}$ with m minimal. We want to show $H_i \in \mathrm{Wall}(C)$. Now let $H_1 = H$.

Let $U = A_{1,H_2} \cap \cdots \cap A_{-1,H_m} \cap H$. We will show: $U \neq \emptyset$ and $\overline{C} \cap H = \overline{U}$.

If $U = \emptyset$, then H doesn't intersect $S = A_{1,H_2} \cap \cdots \cap A_{-1,H_m}$. So either $S \subset A_{1,H}$ or $S \subset A_{-1,H}$. If $S \subset A_{-1,H} \implies S \cap A_{1,H} = \emptyset \implies C = \emptyset$, contradiction. If $S \subset A_{1,H} \implies C = S \cap A_{1,H} = S$, contradicting the minimality. So $U \neq \emptyset$.

Now if $v \in \overline{U} \subset H$. Then for any $\varepsilon > 0$, $B(v, \varepsilon)$ intersects $U \subset S$, and it intersects $A_{1,H}$. So $B(v, \varepsilon) \cap C \neq \emptyset \implies v \in \overline{C} \cap H$. Conversely, if $v \in \overline{C} \cap H$, then for all $\varepsilon > 0$, $B(v, \varepsilon)$ intersects $C \subset S$ and H. So $B(v, \varepsilon)$ intersects $S \cap H = U \implies v \in \overline{U}$.

Fix $C_0 \in \mathcal{C}, C_0 = A_{1,H_1} \cap \cdots \cap A_{1,H_r} \cap A_{-1,H_{r+1}} \cap \cdots \cap A_{-1,H_m}$. We can replace u_{H_i} by u_{-H_i} for all $i > r$, then we may assume

$$C_0 = A_{1,H_1} \cap \cdots \cap A_{1,H_m}; \quad H_i \in \mathrm{Wall}(C_0)$$

Call such a chamber C a **fundamental chamber**. Clearly W acts on $\mathcal{C}$.

Proposition 4.13. *Let $C_0 \subset \mathcal{C}$ be fundamental chamber. Let $I \coloneqq \mathrm{Wall}(C_0), W' = \langle s_H \mid H \in I \rangle \leq W$*

1. *W' acts transitively on $\mathcal{C}$*
2. *$\mathrm{stab}_{W'}(C) = \{1\}, \forall C \in \mathcal{C}$*

Proof. 1. For all $C \in \mathcal{C}, H \in \mathrm{Wall}(C)$, we know $C \sim_H s_H C$. So if $C = wC_0, H \in I$, then

$$\overline{s_H C_0} \cap H = \overline{C_0} \cap H = \overline{U}$$

for $0 \neq U \subset H$. So $\overline{ws_H C_0} \cap wH = w\overline{U}$, and $\overline{C} \cap wH = \overline{wC_0} \cap wH = w\overline{U}$. Hence indeed

$$ws_H C_0 \sim_{wH} wC_0, \quad wH \in \mathrm{Wall}(wC_0)$$

So $w \cdot \mathrm{Wall}(C_0) \simeq \mathrm{Wall}(w \cdot C_0)$

Now we pick a gallery $C_0 \sim C_1 \sim \cdots C_n = C$. We want to inductively show $C_i \in W' \cdot C_0$. The case for $n = 1$ is clear. Now assume $C_i = w_i C_0$ for $w_i \in W'$, and $C_i = w_i C_0 \sim_{H_i} C_{i+1}$, where $H_i \in \mathrm{Wall}(w_i C_0) = w_i \mathrm{Wall}(C_0)$. So we may write $H_i = w_i \cdot H'$ for some $H' \in I$. But we saw that

$$w_i C_0 \sim_{w_i H' = H_i} w_i s_{H'} C_0$$

By uniqueness of adjacency: $w_i s_{H'} C_0 = C_{i+1}$, we are done.

2. Suppose $w \in W', wC_0 = C_0$. Write $w = s_1 \cdots s_d$ for $s_i = s_{H_i}$. We also assume d minimal. Suppose for contradiction $d \geq 1$.

   Let $w_i = s_1 \cdots s_i$, we have a gallery

$$C_0 \sim_{H_1} w_1 C_0 \sim_{w_1 H_2} w_2 C_0 \sim \cdots \sim w_d C_0 = C_0$$

But note we have the following fact: if $C \sim_H D$ and $C \subset A_{-1,H'}$, then either $H = H'$ or $D \subset A_{-1,H'}$.

Now $w_1 C_0 \in A_{-1,H_1}$. We can find some i such that $w_i H_{i+1} = H_1$. So

$$s_{H_1} = w_i s_{H_{i+1}} w_i^{-1} \iff s_2 \cdot s_i = s_1 \cdots s_{i+1}$$

So $w = s_1 \cdots s_d$ is not a reduced word, contradiction.

□

Theorem 4.14. *If C_0 is fundamental chamber, $I = \mathrm{Wall}(C_0)$. Then $W(\mathcal{H}) = \langle s_H \mid H \in I \rangle$. In particular, W acts freely transitively on $\mathcal{C}$.*

Proof. For all $H \in \mathcal{H}$, we will show H is the wall of some chamber $C \in \mathcal{C}$. Once this is done, we can find $w \in W'$ such that $w \cdot C_0 = C$, and $H = w \cdot H'$ for some $H' \in I$. So $s_H = w s_{H'} w^{-1} \in W'$. We are done.

Fix H. Define $\equiv_H$ on $H \backslash \bigcup_{H' \in \mathcal{H} H' \neq H} H' \cap H$ by $x \equiv_H y$ if for any $H' \neq H$, x, y are in the same H'-half space. Now if U is any equivalence class of $\equiv_H$, we write

$$U = A_{1,H_1} \cap \cdots \cap A_{1,H_r} \cap A_{-1,H_{r+1}} \cdots \cap A_{-1,H_m} \cap H$$

Now let S be the set of all half spaces in V containing U. We set $C = \cap_{A \in S} A \cap A_{1,H}$. Note that $U = \cap_{A \in S} A \cap H$, so $\overline{C} \cap H = \overline{U}$. It remains to show C is indeed a chamber.

First of all, C is cut by half spaces. So we only need to any two elements $u, v \in C$ are equivalent. Suppose not, then there exists H' such that $u \in A_{1,H'}, v \in A_{-1,H'}$, say. So $C \nsubseteq A_{1,H'}$ or $A_{-1,H'} \implies A_{1,H'}, A_{-1,H'} \notin S \implies H' \cap U \neq \emptyset$. Contradicting the fact $\equiv_H$ being an equivalence relation. □

4.4 Fundamental Domains

Definition 4.15. Let G acts on set X. We say $D \subset X$ is a **fundamental domain** if for all $x \in X$, $|G \cdot x \cap D| = 1$.

Lemma 4.16. *1. If D is a fundamental domain for G acting on X, then for any $g \in G$*

$$gD \cap D = \{x \in D \mid gx = x\}$$

2. If $\chi = \{gD \mid g \in G\}$, then G acts transitively on χ. Moreover

$$\ker(G \to \mathrm{Sym}(\chi)) = \ker(G \to \mathrm{Sym}(X))$$

Example. *Let $W = W(\mathcal{H})$, and $\mathcal{H}$ be locally finite, W-invariant set of hyperplanes. Let $\mathcal{C}$ be the Coxeter complex, with fundamental chamber $C_0 \in \mathcal{C}$. Then $D \coloneqq \overline{C_0}$ is a fundamental domain for W-action on V.*

Let V be an inner product space, $\mathcal{I}(V)$ is topologized by

$$d(f, g) = \sup_{\|v\|=1} \|f(v) - g(v)\|$$

This gives $\mathcal{I}(V), \mathrm{O}(V)$ a topological group structure. Moreover, for $\mathrm{O}(V)$ the topology agrees with the matrix topology.

Remark. *The topology on $\mathcal{I}(V)$ is determined by open subsets*

$$C(K,U) = \{f \in \mathcal{I}(V) \mid f(K) \subset U\}$$

for K compact, U open. This is the ***compact-open topology****.*

Proposition 4.17. *If $G \leq \mathcal{I}(V)$, and let $C \subset V$ be open and $D = \overline{C}$ is a fundamental domain for the G-action on $\bigcup_{g\in G} g \cdot D$. Then G is discrete.*

Proof. Fix $v \in C$, define $\theta_v : \mathcal{I}(V) \to V, f \mapsto f(v)$. Since θ_v is continuous, we know $C_v = \theta_v^{-1}(C)$ is open in $\mathcal{I}(V)$. We wish to show $C_v \cap G = \{1\}$. This immediately implies G is discrete.

Now pick $g \in C_v \cap G$. So

$$gv \in g \cap gC \subset D \cap gD = \{x \in D \mid gx = x\}$$

Now set $S = \{x \in V \mid gx = x\}$ to be closed in V, and it is in fact isometric to a linear subspace. We see $C \cap gC = C \cap S$ is open in V. This forces $C \leq S$ and $S = V$. Hence $g = 1$ as required. □

4.5 Finite Coxeter Groups

Let (W, I, M) be a Coxeter system, with $(\cdot,\cdot)_M$ positive definite. Let $\sigma : W \to \mathrm{GL}(V_M) = \mathrm{GL}(V)$ be the faithful geometric representation.

For each $i \in I$, set $H_i = \{v \in V \mid (v, e_i) = 0\}$, $A_i = \{v \in V \mid (v, e_i) > 0\}$, so $V = H_i \cup A_i \cup -A_i$. Let $C = \bigcap_{i\in I} A_i$ be open in V, $D = \overline{C}$.

Lemma 4.18. *For all $w \in W, i \in I$, we have*

1. *If $l(x_i w) > l(w)$, then $w \cdot C \subset A_i$*
2. *If $l(x_i w) < l(w)$, then $w \cdot C \subset -A_i$*

Proof. Clearly one implies two. Now if $l(x_i w) > l(w) \implies l(w^{-1}x_i) > l(w^{-1}) \implies w^{-1}e_i > 0 \implies w^{-1}e_i = \sum \alpha_j e_j, \alpha_j \geq 0$. Now if $v \in C \implies (v, e_j) > 0$ for all j, so $(wv, e_i) = (v, w^{-1}e_i) > 0 \implies wv \in A_i$ □

Theorem 4.19. *D is a fundamental chamber for W-action on $\bigcup_{w\in W} w \cdot D$*

Proof. If $wC = C$, then $wC \subset A_i$ for each i. So $l(x_i w) > l(w)$ for each i. This forces $w = 1$. Now by definition we have $|W \cdot x \cap D| \geq 1$ for each $x \in \bigcup_{w\in W} w \cdot D$. Now if $y, z \in W \cdot x \cap D$, then $z = wy$ for some $w \in W$. Hence it suffices to show w fixes y.

Let $J = \{i \in I \mid (y, e_i) = 0\}, K = \{i \in I \mid (z, e_i) = 0\}$. We will prove by induction on $l(w)$ such that $J = K$ and $w \in W_J$. If so, then $(wy, e_i) = (y, w^{-1}e_i)$, which equals (y, e_i) (splittinng into cases $i \in J$ or not). The case for $l(w) = 0$ is trivial.

Now assume $l(w) > 0$. We choose x_i such that $l(x_i w) < l(w) \implies wC \subset -A_i$. So

$$wD = \overline{wC} \subset -\overline{A_i} = -A_i \coprod H_i$$

Hence $z \in D \cap wD \subset H_i \implies i \in K$. Now that $x_i z = z = wy \implies z = (x_i w)y = w'y$ and $l(w') < l(w)$. By induction we know $J = K$, and $w' \in W_J = W_K$. Moreover, $w = x_i w' \in W_K$ as required. □

Corollary 4.20. *If (W, I, M) is a Coxeter system, $|I| < \infty$, and $(\cdot,\cdot)_M$ is positive definite. Then W is finite. Hence the following are equivalent:*

1. *W is finite Coxeter group*
2. *W is positive Coxeter group*
3. *W is finite reflection group*
4. *$W = W(\Phi)$ for some root system Φ*

Proof. Let $D = \overline{C}$. Then W acts on $\bigcup w \cdot D$ with D a fundamental domain. So W is generated by $\{s_{H_i} \mid i \in I\}$. In particular $W \leq \mathrm{O}(V)$ is discrete. But $\mathrm{O}(V)$ is compact. So W is finite. □

5 Affine Coxeter Groups

5.1 Affine Weyl Groups

Let V be an inner product space, $\Phi \subset V$ a crystallographic root system.

For each $\alpha \in \Phi, k \in \mathbb{Z}$,define $H_{\alpha,k} = \{v \in V \mid (v, \alpha) = k\}$ to be an affine hyperplane. Set $s_{\alpha,k} = s_{H_{\alpha,k}} : v \mapsto v - \frac{2(v,\alpha)-k}{(\alpha,\alpha)}\alpha$, and $s_\alpha = s_{\alpha,0}$. Let $\mathcal{H} = \{H_{\alpha,k} \mid \alpha \in \Phi, k \in \mathbb{Z}\}$. Then clearly $\mathcal{H}$ is locally finite and $W(\Phi) \leq W(\mathcal{H})$. We also define the **coroot** of $\alpha \in \Phi : \alpha^\vee = \frac{2\alpha}{(\alpha,\alpha)}$.

Definition 5.1. The coroot lattice is $\mathbb{Z}\Phi^\vee$. Under the isomorphism $V \to T(V)$, we denote the image of $\mathbb{Z}\Phi^\vee$ as $L(\Phi^\vee)$.

Note $W(\Phi)$ normalizes $L(\Phi^\vee)$, since $wT_v w^{-1} = T_{w(v)}$. Therefore we get

$$L(\Phi^\vee) \rtimes W(\Phi) \leq \mathcal{I}(V)$$

Remark. *1. If Φ is not crystallographic, then $\mathbb{Z}\Phi^\vee$ is not discrete (in fact is dense in V)*

2. If Δ is a fundamental system in Φ, then $\Delta^\vee$ is a $\mathbb{Z}$-basis for $\mathbb{Z}\Phi^\vee$.

We let $\widetilde{W(\Phi)} \coloneqq W(\mathcal{H})$, this is called the **affine Weyl group**.

Proposition 5.2. *1. $\widetilde{W(\Phi)} = L(\Phi^\vee) \rtimes W(\Phi)$*

2. $\mathcal{H}$ is W-invariant, and $\widetilde{W(\Phi)}$ is an affine Weyl group.

Proof. 1. Note that $T^k_{\alpha^\vee} \cdot s_\alpha = s_{\alpha,k}$, so $T^k_{\alpha^\vee} \in \widetilde{W(\Phi)}$. This shows $L(\Phi^\vee) \subset \widetilde{W(\Phi)}$. The converse inclusion is also clear from this characterization.

2. We just need to check the following two identites:

$$w \cdot H_{\alpha,k} = H_{w(\alpha),k}$$

$$T_{\beta^\vee}(H_{\alpha,k}) = H_{\alpha,k+(\beta^\vee,\alpha)}$$

□

Example. *If* $V = \mathbb{R}$, $\Phi = \{\frac{1}{2}, -\frac{1}{2}\}$, $\mathcal{H} = \frac{1}{2}\mathbb{Z}$. *Then* $\widetilde{W(\Phi)} = D_\infty = \mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$

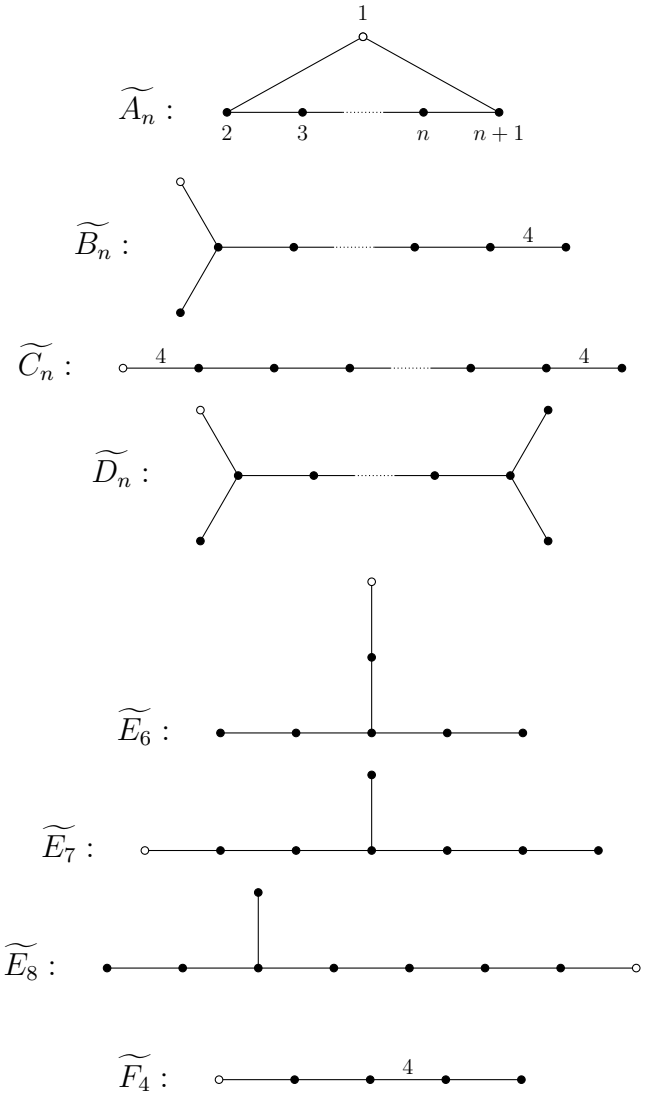
Example. *If* $V = \mathbb{R}^2$, $\Phi = A_2$. *Then* $\widetilde{W(\Phi)} = (\mathbb{Z}\alpha^\vee \oplus \mathbb{Z}\beta^\vee) \rtimes S_3$. *This is of type* $\widetilde{A_2}$.

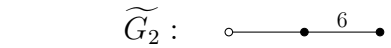
We now record the following fact:

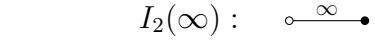
Let $\widetilde{W(\Phi)} = W(\mathcal{H})$ be an affine Weyl group. If $\Delta \subset \Phi$ is a fundamental system, then there exists a chamber in the Coxeter complex whose walls are determined by $H_\delta, \delta \in \Delta$, and $T_{\alpha^\vee} H_\alpha$ for some $\alpha \in \Phi$. So $\widetilde{W(\Phi)} = \langle s_\delta, s_{T_{\alpha^\vee} H_\alpha} \mid \delta \in \Delta \rangle$. Removing any generator, we get a finite group.

5.2 Groups of Positive Type

Let (W, I, M) be a Coxeter system of positive type but not positive. Assume W irreducible, then W has one of the following types:



$\widetilde{G_2}$: 

$I_2(\infty)$: 

Note if Φ has type B_n, then $\Phi^\vee$ is crystallographic of the same type. But $\widetilde{W(\Phi)}, \widetilde{W(\Phi^\vee)}$ have type $\widetilde{B_n}, \widetilde{C_n}$ is some order.

In general, if (W, I, M) has positive type, then $V_M^\perp = \langle \alpha_1 e_1 + \cdots \alpha_n e_n \rangle$ for each $\alpha_i > 0$. If $V_i = \text{Span}\{e_j \mid j \neq i\}, W_i = \langle x_j \mid j \neq i \rangle$ is finite.

Now $\Phi = \{w \cdot e_j \mid w \in W_i, j \neq i\}$ is a root system in V_i, and $W \simeq \widetilde{W(\Phi)}$ via

$$x_j \mapsto s_{e_j}, \quad x_i \mapsto s_{T_{\alpha^\vee} H_\alpha} \text{ for } \alpha = \alpha_1 e_1 + \cdots + \widehat{\alpha_i e_i} + \cdots \alpha_n e_n$$

5.3 The Fundamental Domain

Let $\mathcal{H}$ be a locally finite, $W(\mathcal{H})$-invariant set of hyperplanes in an inner product space V. Let $W = W(\mathcal{H})$. Choose a fundamental chamber C_0 of $\mathcal{C}$. Then $I = \text{Wall}(C_0)$ is a finite set, and W is generated by s_H for $H \in \text{Wall}(C_0)$.

For $H, H' \in I$, let $m_{H,H'}$ be the order of $s_H s_{H'}$. Let $M_{\mathcal{H}} = (m_{H,H'})_{H,H' \in I}$ be the associated Coxeter matrix. Let $(\widetilde{W(H)}, I, M_{\mathcal{H}})$ be the Coxeter system, with generators $x_H, H \in I$. There is clearly a surjection $\theta : \widetilde{W(\mathcal{H})} \to W(\mathcal{H})$.

Theorem 5.3. *For $w \in \widetilde{W(\mathcal{H})}, H \in I$.*

1. *If $l(x_H w) > l(w)$, then $\theta(w)(C_0) \subset A_{1,H}$*
2. *If $l(x_H w) < l(w)$, then $\theta(w)(C_0) \subset A_{-1,H}$*
3. *θ is an isomorphism. In particular, $W(\mathcal{H})$ is a Coxeter group.*

Proof. "2 $\implies$ 3": If $w \neq 1$, then we can pick H such that $l(x_H w) < l(w)$. So $\theta(w)(C_0) \subset A_{-1,H} \implies \theta(w) =\neq 1$. So θ is indeed injective. "1 $\implies$ 2" is clear. Now we show the first statement.

First assume $|I| = 2, \text{Wall}(C_0) = \{H_1, H_2\}$. Now if $H_1 \cap H_2 \neq \emptyset$, we can translate and assume H_1, H_2 are both linear. We're thus done by the corresponding result for finite reflection groups. So we may assume $H_1 \cap H_2 = \emptyset$, then $T = s_{H_2} s_{H_1}$ is translation. Now if $l(x_{H_1} w) > l(w)$, then $\theta(w) = (s_{H_2} s_{H_1})^k = T^k$ or $T^k s_{H_2}$. But in this case indeed $\theta(w)(C_0) \subset A_{1,H_1}$ from the geometric picture.

For the general case, we use induction on $l(w)$. Assume $l(w) > 0$, we pick $H \in I$ such that $l(x_{H'} \cdot w) < l(w)$. Choose $w_0 \in \widetilde{W(\mathcal{H})}$ and $v \in \langle x_H, x_{H'} \rangle$ such that

1. $w = v w_0$
2. $l(w) = l(v) + l(w_0)$
3. $l(w_0)$ is minimal with these properties.

Claim: $l(x_H w_0), l(x_{H'} w_0) > l(w_0)$ and $l(x_H v) > l(v)$

If this claim is true, then by induction hypothesis:

$$\theta(w)(C_0) = \theta(v)(\theta(w_0)(C_0)) \subset \theta(v)(A_{1,H} \cap A_{1,H'}) \subset A_{1,H}$$

by the case of $|I|= 2$. We are therefore done. But the proof of the claim is identical to that appeared in the proof for faithfulness of geometric representation. □

Corollary 5.4. *$D = \overline{C_0}$ is a fundamental domain for $W(\mathcal{H})$-action on V. In particular, $W(\mathcal{H})$ is a discrete subsgroup of $\mathcal{I}(V)$.*

Proof. Firstly, since ever $H \in \mathcal{H}$ is a wall of some chamber, and each chamber is related to the fundamental chamber via a gallery, so we indeed have

$$\bigcup_{w \in W(\mathcal{H})} w \cdot D = V$$

So for each $x \in V$, $|W \cdot x \cap D| \geq 1$. We need to show it's in fact an equality. Now assume $y, z = wy \in W \cdot x \cap D$, we need to show $y = z$.

We induct on $l(w)$. If $l(w) = 0$, then there's nothing to prove. Otherwise, choose $H \in I$ such that $l(s_H w) < l(w) \implies w \cdot C_0 \leq A_{-1,H} \implies w \cdot D = A_{-1,H} \coprod H$. So

$$z = wy \in D \cap wD = H$$

Hence $z = s_H z = s_H wy$, $l(s_H w) < l(w)$. We're therefore done by induction. □

5.4 Infinite Reflection Groups

Proposition 5.5. *If $\bigcap_{H \in \mathrm{Wall}(C_0)} H \neq \emptyset$, then $W(\mathcal{H})$ is finite, and there exists $T \in T(V)$ such that $TW(\mathcal{H})T^{-1}$ is a FRG*

Proof. After translation, we may assume $0 \in \bigcap_{H \in \mathrm{Wall}(C_0)} H$ and hence $W(\mathcal{H})$ is a LRG. Now 0 is fixed by each $s_H, H \in \mathcal{H} \implies x \in \bigcap_{H \in \mathcal{H}} H$. By locally finiteness, we know $\mathcal{H}$ is finite. But $W(\mathcal{H})$ acts freely transitively on $\mathcal{C}$, so $|W(\mathcal{H})|= |\mathcal{C}|$ is finite. It follows that $W(\mathcal{H})$ is a FRG. □

Now let $W = W(\mathcal{H})$, for $\mathcal{H}$ locally finite and $W(\mathcal{H})$-invariant. Fix a fundamental chamber C_0. In light of the previous result we may assume $\bigcap_{H \in \mathrm{Wall}(C_0)} H = \emptyset$. The goal is to show $W(\mathcal{H})$ is an affine Weyl group, i.e. there exists a crystallographic root system Φ such that $W(\mathcal{H}) \simeq \widetilde{W(\Phi)}$

Lemma 5.6. *The map $P : \mathcal{I}(V) \to \mathrm{O}(V)$ given by $\alpha \mapsto T^{-1}_{\alpha(0)}\alpha$ is a surjective group homomorphism, $P^2 = P$, with $\ker P = T(V)$. Hence:*

$$\mathcal{I}(V) = T(V) \rtimes \mathrm{O}(V)$$

Proof. Exercise. □

Now set $\overline{W} = P(W) \leq \mathrm{O}(V)$, $W_x = \{w \in W \mid wx = x\}$ and in particular $W_0 \subset \overline{W} \leq \mathrm{O}(V)$. Let $L(\mathcal{H}) = W(\mathcal{H}) \cap T(V)$, and $\mathcal{H}_l$ be the set of linear hyperplanes in H parallel to some $H' \in \mathcal{H}$.

The agenda is to prove the following: after translation,

1. $W_0 = \overline{W} = W(\Phi)$ for some crystallographic root system Φ
2. $L(\mathcal{H})$ is a lattice in $T(V)$. In fact $L(\mathcal{H}) \simeq L(\Phi^\vee)$
3. $W(\mathcal{H}) = L(\mathcal{H}) \rtimes W_0 = \widetilde{W(\Phi)}$

Lemma 5.7. *$\overline{W}$ is a FRG equals $\langle s_H \mid H \in \mathcal{H}_l \rangle$, and if $s_v \in \overline{W}$ for some $v \in V$, then $H_v \in \mathcal{H}_l$.*

Proof. For each $H \in \mathcal{H}$, let u_H be a normal vector to H. Then $P(s_H) = s_{H'}$ for some $H' \in \mathcal{H}_l$. One thus sees that $\overline{W} = \langle s_{H'} \mid H' \in \mathcal{H}_l \rangle$ is a FRG, since $\mathcal{H}_l$ is a finite, $W(\mathcal{H}_l)$-invariant set. Now if $s_v \in \overline{W}$, say $s_v = P(s_H)$ for some $H \in \mathcal{H}$. One sees $H = T_{S_H(0)} H_v \implies H_v \in \mathcal{H}_l$. $\square$

Now if $U = \mathrm{Span}_{\mathbb{R}}\{u_H \mid H \in \mathcal{H}\} = \mathrm{Span}_{\mathbb{R}}\{u_H \mid H \in \mathcal{H}_l\}$. We know $\overline{W} = W(\Phi)$ for some root system Φ. We choose a fundamental system $\{v_1, \cdots, v_m\}$ of Φ os that it forms a basis for U and $\overline{W} = \langle s_{v_i} \mid 1 \leq i \leq m \rangle$. For each $1 \leq i \leq m$, we choose $H_i \in \mathcal{H}$ such that $H_i \perp v_i$. Let $x \in \cap_{i=1}^m H_i \neq \emptyset$ (exercise). Then we know $\overline{W} = T_x^{-1} \langle s_{H_i} \rangle$

Let $W' = \langle s_{H_i} \rangle$, then we know $T_x^{-1} W' T_x = \overline{W}$. After translation, we may assume $x = 0$ and hence $\overline{W} = W' \subset W_0$. So indeed $W_0 = \overline{W}$. This also shows $\mathcal{H}_l = \{H \in \mathcal{H} \mid H \text{ linear}\}$.

Now since $\bigcap_{H \in \mathrm{Wall}(C_0)} H = \emptyset$, we can find some $H \in \mathrm{Wall}(C_0)$ such that $0 \in H$. But we can choose $H' \in \mathcal{H}_l$ parallel to H. Then $s_H s_{H'}$ is a nonzero translation. Hence $L(\mathcal{H})$ is infinite, so is $W(\mathcal{H})$.

Note $W \to \mathcal{I}(U), \alpha \mapsto \alpha|_U$ is well-defined and injective. We may assume $U = V$.

Proposition 5.8. *The set $L(\mathcal{H})$ is a lattice in $T(V) \simeq V$, and $L(\mathcal{H})$ is generated as an abelian group by*

$$\{s_H s_{H'} \mid H \in \mathcal{H}, H' \in \mathcal{H}_l, H /\!\!/ H'\}$$

and $L(\mathcal{H}) \rtimes \overline{W} = W(\mathcal{H})$

Proof. Since $W(\mathcal{H})$ is a discrete subgroup of $\mathcal{I}(V)$, so $L(\mathcal{H})$ is a discrete subgroup of $T(V)$, hence a lattice.

Now let $L' = \{s_H s_{H'} \mid H \in \mathcal{H}, H' \in \mathcal{H}_l, H /\!\!/ H'\} \subset L(\mathcal{H})$. It suffices to show $W(\mathcal{H}) = L' \rtimes \overline{W}$. Indeed for any $H \in \mathcal{H}$, we pick $H' \in \mathcal{H}_l$ parallel to H. Then $s_H = (s_H s_{H'}) s_{H'} \in L' \rtimes \overline{W}$. $\square$

Theorem 5.9. *$W(\mathcal{H}) = \widetilde{W(\Phi)}$ for certain $\Phi \subset V$ crystallographic root system.*

Proof. For each $H \in \mathcal{H}_l$, let $\lambda_H > 0$ be minimal such that $\lambda_H u_H \in H'$ for some $H' /\!\!/ H, H' \notin \mathcal{H}_l$. Note we can choose λ_H minimal since $L(\mathcal{H})$ is discrete. Let $\alpha_H = \lambda_H u_H$, then $s_{T_{\alpha_H} H} s_H = T_{2\alpha_H}$. By the previous proposition, we know

$$L(\mathcal{H}) = \mathrm{Span}_{\mathbb{Z}}\{T_{2\alpha_H} \mid H \in \mathcal{H}_l\}$$

But $W_0 = \overline{W} = W(\Phi)$ for root system $\Phi = \{\pm\frac{\alpha_H}{\|\alpha_H\|^2} \mid H \in \mathcal{H}_l\}$. We now compute

$$(\frac{\alpha_H}{\|\alpha_H\|^2})^\vee = 2\alpha_H$$

So indeed $L(\mathcal{H}) = L(\Phi^\vee)$, $W(\mathcal{H}) = L(\Phi^\vee) \rtimes W(\Phi)$. It remains to show Φ is crystallographic.

In fact, if $\alpha, \beta \in \Phi^\vee$, then

$$s_\alpha(\beta) = \beta - \frac{2(\alpha,\beta)}{(\alpha,\alpha)}\alpha \in \mathbb{Z}\Phi^\vee \implies \frac{2(\alpha,\beta)}{(\alpha,\alpha)}\alpha \in \mathbb{Z}\Phi^\vee$$

But by minimality of $\alpha = \alpha_H$, we know $\frac{2(\alpha,\beta)}{(\alpha,\alpha)} \in \mathbb{Z} \implies \Phi^\vee$ crystallographic, so Φ is crystallographic. $\square$